

L3 Programming Implementation Rubric

What is a Rubric?

A rubric is a scoring guide used to evaluate performance. In our case, it is used to assess how well you have understood the materials taught.

It has three parts:

1. assessment criteria, grouped by learning objective;
2. rating scale;
3. indicators.

For you and your assessor, the rubric defines what is expected and what will be assessed.

Criteria

Learning Objectives	Assessment Criteria
1. Explain the principles and concepts of programming languages.	1.1 Explain the principles and concepts of programming languages.
2. Understand how programming is implemented.	2.1 Interpret variables within programming languages. 2.2 Interpret common programming control structures that are used when developing code. 2.3 Describe the use of common data structures. 2.4 Describe how algorithms are used in programming. 2.5 Describe how to test and debug programs.
3. Understand the importance of good programming practices.	3.1 Identify the elements of common coding standards. 3.2 Explain the role and importance of good coding practices.
4. Understand functional and non-functional requirements in programming	4.1 Identify the types of functional and non-functional requirements. 4.2 Describe the tests used for functional and non-functional requirements.

Assessments

General information

The assessment will consist of three parts, to be taken in order.

Part 1: 25 Multiple Choice Questions

Part 2: Longform

The long-form questions for this module consist of:

- a runnable example with code to read and run
- questions to fill out (essay style)
- a text box to submit debugged code

Part 3: Viva

This is your opportunity to explain to the assessor how you worked through the problems in the previous tasks.

Important

We recommend that you attempt to answer all the questions even if you are unsure. No negative marks will be taken for wrong answers, however, any unattempted questions will automatically get a score of 0 points.

Grading scheme per task

Part 1: Multiple Choice Questions

Assessment Criteria	Fail	Pass (70%)	Max Grade
Scoring	0 - 17	17.5 - 25	25 (20% of total)
All	Quiz Questions		

Part 2: Longform questions

Assessment Criteria	Fail		Pass (70%)		Max Grade
Scoring	2	5	7	10	100 (80% of total)
1.1 Explain the principles and concepts of programming languages.	1.1 The learner has attempted the flowchart but got two or fewer items correct	1.1 The learner made an attempt at the flowchart, but got fewer than five items correct	1.1 The learner filled in the flowchart in full, with one or two mistakes	1.1 The learner filled in the flowchart correctly	
2.1 Interpret variables within programming languages.	2.1 The learner did not correctly identify and interpret (data type and usage) the	2.1 The learner correctly identified and interpreted at least one variable in the code, but fewer	2.1 The learner correctly identified and interpreted at least four variables in the code	2.1 The learner correctly identified and interpreted all variables	

	variables used in the program	than four			
2.2 Interpret common programming control structures that are used when developing code.	2.2 The learner has not correctly identified and explained the control structures.	2.2 The learner has correctly identified both control structures but has not explained either correctly.	2.2 The learner has correctly identified and explained one of the control structures, but not the other	2.2 The learner has correctly identified and explained both control structures	
2.3 Describe the use of common data structures.	2.3 The learner did not identify the data structure used	2.3 The learner identified the data structure used but did not explain how it was used.	2.3 The learner correctly identified the data structure and explained how it was used	2.3 The learner correctly identified the data structure and which data type it contains and explained how it was used	
2.4 Describe how algorithms are used in programming.	2.4 The learner did not correctly describe how the algorithm was used (less than 2 items correctly described)	2.4 The learner did not correctly describe how the algorithm was used (2-5 parts correctly described)	2.4 The learner correctly described how the algorithm was used.	2.4 The learner correctly described how the algorithm was used in detail (e.g. mentioning data type)	
2.5 Describe how to test and debug programs	2.5 The learner has not correctly identified the bugs in the code. When run, the submitted	2.5 The learner has correctly identified the bugs in the code but has not explained how they	2.5 The learner has correctly identified and explained at least four bugs in the code. The fixed	2.5 The learner has correctly identified and explained all bugs in the code, and the fixed code	

	code still produces error messages	were fixed OR The learner has correctly identified and explained three or fewer bugs in the code When run, the submitted code still produces error messages	code has no more error messages	works correctly	
3.1 Identify the elements of common coding standards.	3.1 The learner did not correctly identify any of the best practices	3.1 The learner identified two best practices correctly	3.1 The learner identified three best practices correctly	3.1 The learner identified more than three best practices	
3.2 Explain the role and importance of good coding practices.	3.2 The learner did not correctly explain the importance of any of the best practices	3.2 The learner explained the importance of two best practices correctly	3.2 The learner explained the importance of three best practices correctly	3.2 The learner explained the importance of more than three best practices correctly	
4.1 Identify the types of functional and non-functional requirements.	4.1 The learner has not correctly identified any requirements	4.1 The learner correctly identified one requirement	4.1 The learner has correctly identified two requirements	4.1 The learner has correctly identified more than two requirements	

4.2 Describe the tests used for functional and non-functional requirements.	4.2 The learner has not correctly described the tests used for their requirements	4.2 The learner has correctly described the tests used for one of their requirements	4.2 The learner has correctly described the tests used for two of their requirements	4.2 The learner has correctly described the tests used for all requirements listed	
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Part 3: Viva

This oral exam is an opportunity to talk through how you approached the MCQ and long-form tasks. It allows the assessor to reflect on your performance and the competencies you have acquired. While it is not marked, it is **mandatory** for you to submit, and you will fail if it is not provided.